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PI Name:	Limoli, Charles Ph.D.		
Project Title:	NSCOR: Mechanisms Underlying Charged Particle-Induced Disruption of CNS Function		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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PI Email:	climoli@uci.edu	**Fax:**	FY (949) 824-3566
PI Organization Type:	UNIVERSITY	**Phone:**	(949) 824-3053
Organization Name:	University of California		
PI Address 1:	Dept. of Radiation Oncology		
PI Address 2:	Medical Sciences I, B149		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-2695	**Congressional District:**	45
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Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Acharya, Munjal Ph.D. (University of California, Irvine) Baulch, Janet Ph.D. (University of California, Irvine) Britten, Richard Ph.D. (Eastern Virginia Medical School) Parihar, Vipan Ph.D. (University of California, Irvine) Soltesz, Ivan Ph.D. (University of California, Irvine)		
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Task Description:	Humans exposed to the space radiation environment encounter a different type of radiation than that found on Earth. Highly energetic particles can traverse the body, including the brain, and leave a path of damage that is difficult to repair and disrupts function. One consequence of this exposure is the development of radiation-induced cognitive dysfunction. The focus of this application is to resolve the mechanisms leading to radiation-induced cognitive dysfunction, in order to improve risk estimates for early and late CNS (central nervous system) effects caused by exposure to the space radiation environment. We have found that radiation impacts excitatory and inhibitory circuitry in the brain differently, and leads to significant disruptions to the structural integrity of neurons while altering the levels and activity of the proteins that regulate neuronal function. The situation is further complicated by the fact that different regions of the brain exhibit differential sensitivity to charged particle irradiation. Based on the foregoing we propose a comprehensive series of mechanistic studies analyzing the biochemical, structural, and electrophysiologic changes induced by radiation exposure that alter function to impact cognition. Carefully selected cognitive tasks and a wide variety of approaches will be used to critically analyze microcircuits in the brain and how charged particle-induced changes are transmitted via unique signaling pathways between cells that can alter gene expression profiles within cells. Collectively these studies will span a radiation effects pathway from synapse to cognition, with an overarching goal of determining whether charged particle irradiation leads to alterations in the excitatory and inhibitory activity of the brain at a global and/or region specific level. Our studies will therefore elucidate the functional consequences of radiation-induced changes on cognition that impose risk on mission critical activities and long-term cognitive health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work will characterize several of the adverse effects of ionizing radiation on the CNS and help to define the early and late CNS risks associated with exposure to the space radiation environment. There is a real possibility that many of the consequences of cosmic radiation exposure will extend over many years and impact (possibly negatively) neurocognitive health after return to Earth. Thus, research elucidating the mechanistic basis of charged particle-induced effects in the irradiated brain may well have functional importance to terrestrial situations where the public may be exposed to ionizing radiation from a variety of potential sources.
	Research Highlights for the Central Nervous System NASA Specialized Center of Research (CNS NSCOR): Work under the NSCOR has been able to demonstrate direct links between charged particle induced disruptions to cognition and persistent damage to specific neuronal subsets interrogated by our behavioral tasks. In addition to published findings, we have now uncovered many fascinating effects of single ion, GCR (galactic cosmic radiation), and low dose rate exposures that point further to the elevated risks associated with deep space radiation exposure. Low dose and low dose rate neutron studies: • Low dose rate studies at the Colorado State University (CSU) neutron irradiation facility have been initiated and completed, and resulted in the following publication: Acharya, M.M., Baulch, J.E., Klein, P.M., Baddour, A.A.D., Apodaca, L.A., Kramár, E.A., Alikhani, L., Garcia Jr. C., Angulo, M.C., Batra, R.S., Fallgren, C.M., Borak, T.B., Stark, C.E.L., Wood, M.A., Britten, R.A., Soltesz, I. and Limoli, C.L. New concerns for neurocognitive function during deep space exposures to chronic, low dose rate, neutron radiation. eNeuro, 22;6(4) (2019). pii: ENEURO.0094-19, doi: 10.1523/ENEURO.0094-19 In brief, these findings documented that chronic (6 month) low dose (18 cGy) and dose rate (1 mGy/day) exposure of mice to a mixed field of neutrons and photons resulted in diminished hippocampal neuronal excitability and disrupted hippocampal and cortical long-term potentiation. Furthermore, mice displayed severe impairments in learning and memory, and the emergence of distress behaviors. Behavioral analyses showed an alarming increase in risk associated with these realistic simulations, revealing for the first time, some unexpected potential problems associated with deep space travel on all levels of neurological function. • This study prompted an editorial and a published response to that editorial: Limoli, C.L., Britten, R.A., Baulch, J.E and Borak, T.B. Response to the Commentary from Bevelacqua et al. eNeuro, (2019). pii: ENEURO.0439-19.2019. doi: 10.1523/ENEURO.0439-19. • A manuscript detailing the marked impact of sleep fragmentation in combination with low dose rate neutron exposures has now been published: Britten, R.A., Duncan, V.D., Fesshaye, A., Wellman, L., Fallgren, C.M. and Sanford, L.D. Sleep fragmentation exacerbates executive function impairments induced by protracted low dose rate neutron exposure. Int. J. Radiat Biol. (2019), Dec 6:1-11. doi: 10.1080/09553002.2019.1694190. In brief, this study reports that chronic low dose (18 cGy) and dose-rate (1 mGy/day) exposure of rats to mixed neutron and photon over the course of six months resulted in significant impairment of simple discrimination (SD) performance. Should similar effects occur in astronauts subjected to low dose rate exposure to Space Radiation, the impairment of SD performance would result in a decreased ability to identify and learn the ‘rules’ required to respond to a new task or situation. Analysis of the behavioral data by kernel density estimation revealed that 40% of rats had severe attentional set shifting (ATSET) impairments. This value may be a best-case scenario because exposure to neutrons also adversely impacted performance in the unconstrained cognitive flexibility (UCFlex) task. Furthermore, when the good performing rats were reevaluated after they had been subjected to sleep fragmentation, additional ATSET performance decrements were observed in the set shifting stages of the ATSET test, with only 7.4% of the neutron exposed rats able to successfully perform ATSET under normal and sleep fragmented conditions, as opposed to ~55% of shams. Overall, protracted low dose and low dose rate neutron exposures impairs executive functions in a high percentage of rats that were normally rested; however, further detriments in performance become evident when the rats are subjected to sleep fragmentation. Low dose, single ion studies: • A manuscript detailing the benefits of microglial elimination for the preservation of neurocognitive function after cosmic radiation exposure has now been published: Allen, B.D., Amber R. Syage, A.R., Maroso, M., Baddour, A-A.D., Luong, V., Minasyan, H., Giedzinski, E., West,

Task Progress:

B.L., Soltesz, I., Limoli, C.L., Baulch, J.E and, Acharya, M.M. Mitigation of 4He irradiation-induced brain injury by microglia depletion. J. Neuroinflammation. 2020 May. Accepted and in press. DOI: 10.1186/s12974-020-01790-9 (assigned although not yet online)

In brief, these studies showed that PLX5622 treatment caused a rapid and near complete elimination of microglia in the brain within 3 days of treatment. Irradiated animals on a normal diet exhibited a range of behavioral deficits involving the medial pre-frontal cortex and hippocampus and increased microglial activation. Animals on PLX5622 diet exhibited no radiation-induced cognitive deficits, and expression of resting and activated microglia were almost completely abolished throughout the brain. While PLX5622 treatment was found to attenuate radiation-induced increases in post-synaptic density protein 95 (PSD-95) puncta and to preserve mushroom type spine densities, other morphologic features of neurons and electrophysiologic measures of intrinsic excitability were relatively unaffected. These data suggest that microglia play a critical role in cosmic radiation-induced cognitive deficits in mice and that approaches targeting microglial function may provide attractive candidates for radiation countermeasures.

• A manuscript detailing the different neurological responses between male and female mice following helium ion irradiation is under secondary review:

Parihar, V.K., Angulo, M.C., Allen, B.D., Syage, A., Usmani, M.T., Chapelle, E.P. d-l., Amin, A.N., Flores, L., Lin, X., Giedzinski, E. and Limoli, C.L. Sex Specific Cognitive Deficits Following Space Radiation Exposure. Submitted Front. Behav. Neurosci. (2020).

In brief, these studies showed that female mice exhibit significant resistance to adverse neurocognitive effects of space radiation. Male mice exposed to low doses (= 30 cGy) of energetic (400 MeV/n) helium ions (4He) had significantly higher levels of neuroinflammation and more extensive cognitive deficits than females. Twelve weeks following 4He ion exposure irradiated male mice demonstrated significant deficits in object and place recognition memory accompanied by activation of microglia, marked upregulation of hippocampal Toll-like receptor 4 (TLR4), and increased expression of the pro inflammatory marker high mobility group box 1 protein (HMGB1). Additionally, we determined that as opposed to female cohorts, exposure to 4He ions caused a significant decline of dendritic spine density along hippocampal neurons in male mice. These data indicate that fundamental differences in inflammatory cascades between male and female mice may drive divergent CNS radiation responses that differentially impact the structural plasticity of neurons and neurocognitive outcomes following cosmic radiation exposure.

• A manuscript detailing the cognitive and electrophysiological responses following silicon ion irradiation has now been published:

Britten, R.A., Duncan, V.D., Fesshaye, A., Rudobeck, E., Nelson, G.A. and Vlkolinsky, R. Altered cognitive flexibility and synaptic plasticity in the rat prefrontal cortex following exposure to low (=15 cGy) doses of 28Si particles. Radiat Res. 2020 Mar;193(3):223-235. doi: 10.1667/RR15458.1.

Briefly, this study has established the impact that 1-15 cGy 600 MeV/n 28Si radiation had on cognitive flexibility performance, glutamatergic synaptic transmission and plasticity in the prelimbic area (PrL) of the medial prefrontal cortex (mPFC) of ~10-month-old (at the time of irradiation) male Wistar rats. Exposure to 1 cGy 600 MeV/n 28Si ions resulted in significantly impaired performance in the simple (SD) and compound discrimination (CD) stages of the attentional set shifting (ATSET) task. However, there was a pronounced non-linear dose response for cognitive impairment. Should similar effects occur in astronauts, the impairment of SD performance would result in a decreased ability to identify and learn the "rules" required to respond to new tasks/situations, while the impaired CD performance would result in a decreased ability to identify and maintain focus on relevant aspects of the task being conducted. The irradiated rats were also screened for performance in a task for unconstrained cognitive flexibility (UCFlex), often referred to as creative problem solving. Exposure to 1, 5, and 10 cGy resulted in a significant reduction in UCFlex performance, in an apparent all-or-none responsive manner. Importantly, performance in the ATSET test was not indicative of UCFlex performance. From a risk assessment perspective, these findings suggest that a value based on a single behavioral end point may not fully represent the cognitive deficits induced by space radiation, even within the cognitive flexibility domain. After completion of the cognitive flexibility testing, in vitro electrophysiological assessments of glutamatergic synaptic transmission and plasticity were performed in slices of the PrL cortex of 10 cGy irradiated rats. Extracellular recordings of field excitatory postsynaptic potentials revealed that radiation significantly decreased long-term depression in layer L5. Patch-clamp whole cell recordings in pyramidal neurons of the L2-3 revealed reduced frequency of spontaneous excitatory postsynaptic currents indicating alterations in presynaptic glutamate release and impaired neuronal spiking (e.g., decreased action potential amplitudes) in irradiated neurons. However, there was no obvious correlation between magnitudes of these electrophysiological decrements and the cognitive performance status of the irradiated rats. These data suggest that while radiation-induced changes in synaptic plasticity in the PrL cortex may be associated with cognitive impairment, they are most likely not the sole determinant of the incidence and severity of such impairments.

• Pion exposures: We have now completed our first run exposing male mice to negative pions at the Paul Scherrer Institute (PSI) in Zurich, Switzerland. Data obtained at the first (2 month) and second (5 month) post-irradiation times have revealed a significant behavioral decrement, showing that the pion irradiated mice fail to learn the Object Location Memory/Updating task on the second latter time point. No such deficits were found on the Novel Object Recognition task or the Light/Dark Box test. It should be noted that mice were only exposed to a total dose of 5 cGy of negative pions. Follow up studies are planned for the winter of 2020.

• Our team has coordinated the successful irradiation of a 600-mouse cohort to the month-long, full spectrum complex 33-beam GCR simulation. Several aspects of this major endeavor are worth highlighting:

• The animals have been distributed to 7 groups outside of the home institution at UCI (University of California Irvine).

i. At Harvard, the laboratories of Drs. David Sinclair and Raj Desai are engaged in a series of complementary behavioral analyses implementing, open field testing, aggression testing, endurance/stamina measures and touch screen measures of executive function. Dr. Desai is also performing microdialysis of neurotransmitter levels.

ii. At Northern Illinois, the laboratory of Dr. Douglas Wallace is engaged in testing cohorts for changes in vestibular endpoints. His measures (rope pulling, ocular tracking, balance) provide information relevant to equilibrioception, or if/how GCR exposure may (or may not) disrupt the coordinated interactions between the visual (eye) and vestibular (inner ear) systems.

iii. At Stanford, the laboratory of Dr. Ivan Soltesz is engaged in a variety of paired and network levels electrophysiological recordings designed to evaluate critical functional outcomes that directly mediate neurotransmission. RNAseq analysis and 3D-video behavioral measurements add innovative angles at further assessing the risks of cosmic radiation exposure.

iv. At Eastern Virginia Medical School (EVMS), the laboratory of Dr. Richard Britten continues to optimize state-of-the-art behavioral testing paradigms using rats. His studies evaluate a different species aimed at analyzing changes in executive function (ATTSET), risk taking (gambling), problem solving (UCFlex), and how these measures are impacted by sleep fragmentation. The addition of Dr. Britten's Kernel density estimation and numbers needed to harm (NNH) algorithms provide the NSCOR and NASA with a unique suite of behavioral and analysis tools unmatched by any other group in the world. v. At the Uniformed Services University of the Health Sciences (USUHS), the laboratory of Dr. Dara Dickstein is engaged in performing ultrastructural analysis of neuronal structure and myelination by transmission electron microscopy.

vi. At Indiana University, the laboratory of Dr. John Beggs is undertaking a unique approach aimed at analyzing network connectivity using a microelectrode array to analyze the spatial interactions between multiple cells in efforts to test the theory of "Criticality" or how neuronal ensembles undergo phase transitions, which is hypothesized to enhance information processing.

• At UCI, animals have been distributed to 2 groups, in addition to the concerted efforts by the Limoli laboratory.

i. Dr. Limoli's team continues to analyze the behavioral consequences of cosmic radiation exposure using their well-developed suite of behavioral platforms that now include: Novel Object Recognition (NOR), Object in Place (OiP), Temporal Order (TO), Light/Dark Box (LDB), Forced Swim Test (FST), Fear Extinction (FE), and a newly developed paradigm showing strong cross-species correlates termed Object Location Memory(OLM)/Updating task – which is a more rigorous version of the classical Novel Place Recognition (NPR) task. Additional work carried out in support of the NSCOR includes a wealth of molecular and biochemical analyses aimed at determining the causes of behavioral and electrophysiological deficits.

ii. At UCI, the laboratory of Marcelo Wood facilitates 2 main efforts. The first involves network measurements of Long-Term Potentiation (LTP) in different regions of the brain (hippocampus, medial prefrontal cortex). This measure of synaptic plasticity is thought to facilitate learning and memory. The second involves the mechanisms of memory consolidation, and is focused on using LTP as a tool to probe the basis of epigenetic changes involving histone acetylation, and how drugs that target specific histone deacetylases (HDAC) can resolve radiation-induced and other forms of dementia.

iii. At UCI, the laboratory of Dr. Craig Stark is a behavioral neuroscientist focused on the analysis of structural and functional imaging approaches to evaluate human subjects. He will assist in computation models of cognition, and assist in the analysis of MRI imaging data sets, to pinpoint regions of high/low radiation sensitivity in a non-invasive manner.

• Lastly studies initiated during the complex 33-beam GCR sim included a proprietary compound capable of minimizing the effects of radiation and aging on the CNS as well as other systemic organ targets. This compound termed nicotinamide mononucleotide (NMN) was developed by Dr. David Sinclair at Harvard, and was included in many (not all) of the studies detailed above, and was evaluated by multiple investigators (Sinclair, Soltesz, Limoli) as a potential countermeasure aimed the mitigating the adverse effects of cosmic radiation exposure on the CNS. Final analyses of our findings using the NMN countermeasure are ongoing and will be continue over the coming months.

• Data sets involving all of the aforementioned 10 laboratories are at various stages of completion.

In summary, much of our ongoing studies have added to our knowledge of space radiation effects on the brain. We have now completed some extensive studies utilizing the full spectrum "33-mixed beam" GCR simulation, in addition to the 6-month protracted low dose/low dose rate neutron study at Colorado State University. Initial data sets have been finalized after the first Pion run at the PSI. As alluded to above, a wealth of other studies (including the NMN countermeasure) and more comprehensive data analysis are ongoing and are planned for the upcoming funding period.

Bibliography Type:	Description: (Last Updated: 01/14/2026)
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